SYNOPSIS REPORT ON MINI PROJECT

Title of the Project

“Tomato – Local Restaurants in One Place”

SUBMITTED BY

“Mr. Katate Swapnil Chandrakant”

UNDER THE GUIDANCE OF

“Prof .Vidya Ugare ”

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1.Title of the project :-

- ❖ Tomato – Local Restaurants in One Place

2.Introduction to the project :-

The *Tomato* web application is an online food discovery and ordering platform that brings multiple local restaurants together in one place. Users can explore nearby restaurants, browse menus, add food items to their cart, and place online orders conveniently. The platform aims to simplify the food ordering process while offering restaurant owners an efficient digital presence. The system also integrates a machine learning model that suggests personalized food recommendations to users based on their preferences and past orders.

1. **Objectives :-**

    - To create a user-friendly web platform for exploring and ordering food from local restaurants.
    - To provide secure authentication and authorization for users and admins using JWT (JSON Web Token).
    - To integrate a recommendation system using a machine learning model built in Python.
    - To allow real-time order tracking from restaurant to customer.
    - To enable admins to manage products, menus, and track orders efficiently.
    - To support a seamless payment gateway for online transactions.

2. Scope of the project :-

The *Tomato* project is designed to serve as a comprehensive solution for food ordering and restaurant management. It covers :

- User registration, login, and secure access.
- Dynamic menu display and easy cart management.
- Online payment processing and order tracking.
- Admin dashboard for managing restaurants, products, and deliveries.
- Machine learning-based recommendation engine for personalized user experience.
  The system can be scaled to accommodate multiple restaurants and extended with features such as delivery partner apps, reviews, and loyalty programs.

3. Technologies Used :-

- **Frontend:** React.js , HTML, CSS, JavaScript
- **Backend Framework: Node.js ,express.js,** Python
- **Database:** Mongodb
- **Machine Learning Model:** Python (using Pandas, Scikit-learn)
- **Other Tools:** REST APIs,JavaScript,Vscode

4. Modules of the System :-

1. User Authorization Module:
   Handles login, signup, and authentication using JWT for both users and admins.
2. Menu Display Module :
   Displays restaurant menus dynamically with categories and item details.
3. Cart System Module :
   Allows users to add, remove, and update food items in the shopping cart.
4. Payment Gateway Module :
   Processes secure online payments for food orders.
5. Admin Dashboard Module :
   Enables admin to add, edit, or remove menu items, manage orders, and view analytics.
6. Product Listing Module :
   Displays all available food items with filtering and sorting options.
7. Order Tracking Module:
   Tracks delivery status in real-time using live updates.
8. Machine Learning Recommendation Module :
   Suggests restaurants or food items to users based on behavior and preferences.

5. System Architecture :-

The system follows the **MERN stack architecture**:

- **Frontend (React.js)** communicates with **Backend (Node.js + Express.js)** via REST APIs.
- **MongoDB** stores all data such as users, menus, orders, and transactions.
- **JWT** ensures secure authentication and data flow between client and server.
- **Python ML model** runs as a microservice, providing recommendation data to the backend through API integration.
  This architecture ensures scalability, modularity, and high performance for both users and administrators.

6. **<u>Expected Outcome</u> :-**

- A fully functional and responsive web application for local restaurant discovery and online food ordering.
- Real-time order tracking and seamless user experience.
- Intelligent food recommendation system for users.
- Secure admin dashboard for restaurant management and analytics.
- Increased visibility for local restaurants and customer engagement.

<u>Advantages</u> :-

- Centralized platform for multiple restaurants.
- Secure login and data protection through JWT.
- Personalized user recommendations powered by machine learning.
- Convenient online ordering and tracking system.
- Admin control for efficient restaurant and delivery management.
- Scalable and maintainable MERN-based architecture.

7. **<u>Conclusion</u> :-**

The *Tomato – Local Restaurants in One Place* project effectively integrates modern web technologies and machine learning to provide an intelligent, secure, and user-friendly food ordering experience. By combining React.js, Node.js, Express.js, MongoDB, and Python, the system offers an end-to-end solution for users, restaurant owners, and delivery tracking. This project demonstrates the practical application of full-stack and AI technologies to improve real-world food delivery and restaurant management systems.